

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element F Initial Template - AT - 3 points

How to use this template

Students: The template below is one way to assess whether your proposed solution is viable and realistic, but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit, be sure to delete the instructions.

Element F: Consideration of Design Viability

Judgment about Design Viability (F1)

"Design viability" is the capacity of the proposed solution to solve the problem. Design viability refers to how your design decisions are tethered to your design requirements. For example, if one of your design requirements is that the total cost of the prototype cannot exceed X, then you could present a table of costs for each component of your proposed design solution as justification for why your design is likely to meet that requirement.

"Credible evidence" means it shouldn't just be your hunch that this will work. Be creative about how you validate the viability of your design. You should go beyond just having a stakeholder or expert reviewer say that the design seems viable. For example, you could point to evidence that a sub-system already functions, or you demo a proof of concept, or you've discussed with an expert that said, "I've seen something similar work in this other related situation...", etc.

Use as many rows as needed. Remember, this chart below may help you organize your ideas, but you will need to actually execute the "credible evidence."

Design Viability Description	Connection to Design Requirement from Element C	Credible Evidence?
<i>Example: The solution is</i>	<i>Example: Design does not</i>	<i>Example: Create a table of</i>

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inexpensive	cost more than \$X	costs for each component of the solution.
The solution is a light system	Lighting up the field so athletes can see (high priority)	We found that we would need 4-6 light poles 50-70 ft high to light up areas around the field and on the field and track directly. (sports-field-lighting-standards-layouts-complete-guide)
The system has Light Barriers	Light barriers to stop the lights from shining in the houses (high priority)	The barriers need to be at least as tall or taller than the the actual lights poles itself to 70-100 feet high and need to fit around the track and the barrier must run the length and width of the field, extending along the perimeter closest to the affected residential area (how-to-reduce-spill-light-from-stadium-lighting-for-nearby-residents)
Our system has angled Lights	Angling the lights would provide extra safety for the light not getting into other peoples homes (high priority)	The light poles would have to be aimed inward to create a 25°+ elevation angle from the field center to minimize glare (football-field-lighting-design-stadium-light-factory.html)
Cost	Cost is also a priority because you would not be able to find quality field lights for cheap	Installing 4–6 high-quality LED field lights (poles and fixtures) for a football field with a track typically costs between \$170,000 and \$880,000, with a common range for high school

